

RoSE: Robust Analog Circuit Parameter Optimization with Sampling-Efficient Reinforcement Learning

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Abstract—Design automation of analog circuits has been a long-standing challenge in the integrated circuit field. Recently, multiple methods based on learning or optimization have demonstrated great promise in automating device sizing for analog circuits. However, they often ignore the strong susceptibility of analog circuits to process, voltage, and temperature (PVT) variations or suffer from low sampling efficiency to train algorithms. To address these critical limitations, this paper proposes RoSE, the first Robust analog circuit parameter optimization framework with high Sampling Efficiency by synergistically combining Bayesian Optimization (BO) and reinforcement learning (RL). Its core is to use the fast convergence of BO to find an optimized starting point for the backbone RL agent to notably improve its sampling efficiency during the learning process. With this pre-optimization, we further leverage the RL's superior optimization ability to achieve robust device sizing by incorporating sufficient features of PVT variations into the representation learning loop. Experimental results of our proposed method on exemplary circuits show $3.25\times\sim 16\times$ improvement of sampling efficiency and $6.8\times\sim 24\times$ improvement of figure-of-merit (FoM, defined with design efficiency and design accuracy) as compared to prior methods.

I. INTRODUCTION

Since the advent of integrated circuits (ICs), labor-intensive manual efforts dominate the entire analog circuit design process. Despite its popularity, manual design is faced with two severe drawbacks [1]. First, it requires many iterative procedures to arrive at a reliable design, which is quite time-consuming and unable to fulfill the exponentially-growing demands of the fast-developing IC industry. Second, as technology nodes continuously scale down, the flow becomes more cost-ineffective when being applied to port the same design from one technology node to a more advanced one. To tackle these challenges, the design automation of analog circuits has thus been a long-sought-after goal.

In the past decades, intensive research efforts have been paid to automate the design of analog circuits at the pre-layout stage, where the primary task is to solve the device sizing problem, i.e., finding the optimal device parameters to achieve the given circuit specifications. The early explorations focus on optimization-based methods, e.g., Geometric Programming [2], Genetic Algorithm [3], and Bayesian Optimization [4]. Recently, learning-based methods, such as supervised learning methods [5], [6] and reinforcement learning methods [7]–[12], have emerged as promising alternatives. Despite the great potential, these methods are limited to optimizing device parameters at the nominal condition, causing significant performance degeneration of circuits in practice due to process, voltage, and temperature (PVT) variations. Performance robustness against the PVT variations is of paramount importance to designing analog circuits.

The practical design automation method thus must account for the strong susceptibility of analog circuits to PVT variations to guarantee design reliability. Additionally, although learning-based methods with well-trained models [7]–[11] often demonstrate higher efficacy than optimization-based methods, they suffer from low sampling efficiency as abundant data points from time-consuming circuit-level simulations are required to reach a good learning performance, which could significantly prolong the training process. Developing sampling-efficient learning-based methods can notably facilitate the training-to-deployment loop.

To simultaneously tackle the above two critical challenges, in this paper, we propose RoSE, the first Robust analog circuit parameter optimization framework with high Sampling Efficiency. Particularly, deep reinforcement learning (RL) is taken as the backbone of our framework as it has shown great promise in solving many complex optimization problems (circuit design with PVT variations is such a problem). To improve the sampling efficiency for RL training, we adopt Bayesian Optimization (BO) as a vanguard to find an optimized searching starting point for our RL agent, based on which the agent can be well-trained with much fewer data points. To deliver superior robustness, we carefully design a variation-aware reward function as the optimal objective and delicately select the key domain knowledge of PVT variations as the essential observations. In this way, the RL backbone can directly incorporate PVT variations into its optimization loop, thereby efficiently learning a robust policy to automate the device parameter optimization for analog circuit design. Key contributions to the work are as follows:

- This work presents the first sampling-efficient and robust framework by combining BO and RL to automate the analog circuit device parameter optimization.
- We leverage BO for pre-searching to improve the sampling efficiency of the RL backbone. Together with an efficient batch simulation manner of SPICE, the RL training can be notably expedited.
- We tailor a variation-aware reward function as an optimal objective by weighting sub-rewards from all PVT corners and propose a circuit-aware multi-modal policy network to sufficiently fuse the most essential observations related to PVT variations to enable a robust RL agent.
- Experiments show $3.25\times\sim 16\times$ improvement of sampling efficiency and $6.8\times\sim 24\times$ improvement of figure-of-merit in terms of design efficiency and design accuracy of our method as compared to prior arts.

II. BACKGROUND AND RELATED WORK

A. Design Considerations of Robust Analog Circuits

To ensure the reliability of analog circuits under different harsh environments, the key design consideration is to minimize

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the influence of process (P), voltage (V), and temperature (T), i.e., PVT variations. Process variation depicts the deviation of fabricated devices from an ideal specification due to manufacturing errors. It often includes typical N-type transistor/typical P-type transistor (TT), fast N-type transistor/fast P-type transistor (FF), slow N-type transistor/slow P-type transistor (SS), slow N-type transistor/fast P-type transistor (SF), and vice versa (FS). Voltage/temperature variations are from uncertain ambient changes. The typical environmental temperature range for a circuit's operation is $[-40, 125]^{\circ}\text{C}$. And it is common for the supply voltage to experience a deviation of $\pm 10\%$ from its nominal value, V_{DD} . A single PVT corner is a combination of P, V, and T from their varying ranges. All these variations are unavoidable and can lead to the performance degeneration of a circuit compared to its nominal case, i.e., $\{\text{TT}, V_{\text{DD}}, 25^{\circ}\text{C}\}$.

The inset in the environment of Fig. 1 illustrates a conceptual example of the performance variation of an operational amplifier under different PVT corners. The variation range can be huge across all PVT corners, making the robust design of analog circuits a challenging task. In practice, human experts first carefully choose an initial group of device parameters as the starting point based on the given circuit topology. Then, they use multiple fine-tuning steps to search for robust device parameters to achieve the design goal. With lots of such iterations, only the device parameters that pass the tests under all PVT variations will be taken as the optimal solution. This manual design process is obviously quite time-consuming and cost-ineffective. An efficient and economic design automation method is thus highly desirable.

B. Related Work

Diverse design automation methods have been proposed to tackle the device sizing problem of analog circuit design, such as optimization-based methods and learning-based methods. Compared to optimization-based methods, e.g., Geometric Programming [2], Genetic Algorithm [3], and Bayesian Optimization [4], learning-based methods [5]–[11], e.g., reinforcement learning (RL) methods and supervised learning (SL) methods usually achieve higher design efficiency and design accuracy once their models are well-trained. Especially, recent research advances have shown the superiority of RL methods [7]–[11] to SL methods [5], [6] which often suffer from inherent approximation errors. Instead of learning an input-output mapping with the one-step prediction of SL, RL methods learn a decision policy to update device parameters via multi-step fine deployment to achieve better quality. RL is exceptional in solving many complex optimization problems, such as gaming, and robots, making it an ideal candidate to tackle the complicated circuit design problem with strong PVT variations. We thus choose it as our backbone.

Nonetheless, previous work [2]–[10] tend to ignore PVT variations and are only effective in nominal cases. An automated design method to realize robust sizing is of great importance to explore. To the best of our knowledge, only one prior work, RobustAnalog [11] targets reliable sizing. It uses part of PVT corners for training, limiting the performance and converging speed. Additionally, current learning-based methods [5]–[11] are faced with another critical issue, i.e., low sampling efficiency to train excellent models. Time-consuming circuit-level simulations are required to collect the data points, which can notably delay the training process. Sampling-efficient methods are thus also

highly desired. Our proposed method tackles these two critical challenges and can achieve strong robustness while remaining high sampling efficiency.

III. PROPOSED ROSE FRAMEWORK

A. Problem Formulation

We target the challenging device sizing problem with a given circuit under stringent PVT variations, formulated as

$$\begin{aligned} \min_{s,g} \quad & f(s,g), \\ \text{s.t.} \quad & s = F(x), \\ \text{where} \quad & s \in \mathbb{R}^{i \times j}, \quad x \in S_P, \quad g \in S_G. \end{aligned} \quad (1)$$

Here, the function $f(s,g)$ represents the difference between the circuit specifications s and the design goal g . For example, for operational amplifiers (Op-Amps), there are four main circuit specifications, i.e., gain (G), power consumption (P), phase margin (PM), and bandwidth (BW). $F(\cdot)$ is a circuit simulator environment to get the circuit specifications s based on a set of device parameters x , e.g., width and finger numbers of transistors. s is essentially a matrix where i represents the specification type (e.g., gain) and j represents a PVT corner. Therefore, $s \in \mathbb{R}^{4 \times 16}$, assuming 16 PVT corners for the design of Op-Amps. The device parameter set x is constrained by the design space S_P . The design goal g is restricted to a reasonable sampling space S_G that the circuit can achieve. Our objective is to minimize $f(s,g)$ by efficiently seeking a group of optimal device parameters so that the circuit specifications can meet an arbitrarily given group of design goals under all PVT variations.

B. Framework Implementation

We explore synergistically combining Bayesian Optimization (BO) and reinforcement learning (RL) to achieve sampling-efficient robust sizing. Fig. 1 shows the overview of our framework which contains two parts: a BO Vanguard and an RL Backbone. BO is a well-known optimization algorithm that can often achieve the fastest convergence [4], [9] to an optimum for a given design goal as compared to other optimization techniques [2], [3]. However, it needs to restart from scratch if the given design goal is changed. On the other hand, well-trained RL methods can reach arbitrary design goals without retraining based on a deployment trajectory with a starting point. Unfortunately, for robust analog circuit design—a more complex problem, RL methods demand more data points from more time-consuming circuit-level simulations (i.e., PVT simulations) to sufficiently explore the design space, thereby leading to a low sampling efficiency toward the convergence. With this key insight in mind, we propose to leverage BO as a vanguard to first coarsely seek an optimized starting searching point (i.e., initial device parameters for searching) for our RL agent. On this basis, the RL agent can then be trained with much fewer interactions with time-ineffective circuit-level simulations, thereby improving the sampling efficiency. As conceptually shown in Fig. 1 (the BO Vanguard part), an optimized starting point can help RL to find a shorter trajectory to reach the design goal as compared to the randomly picked-up one. Therefore, it can guide the RL algorithm to converge fast with less training data. Note that BO is only required once in our framework if the sampling space of design goals is defined/given. The optimized starting point is then

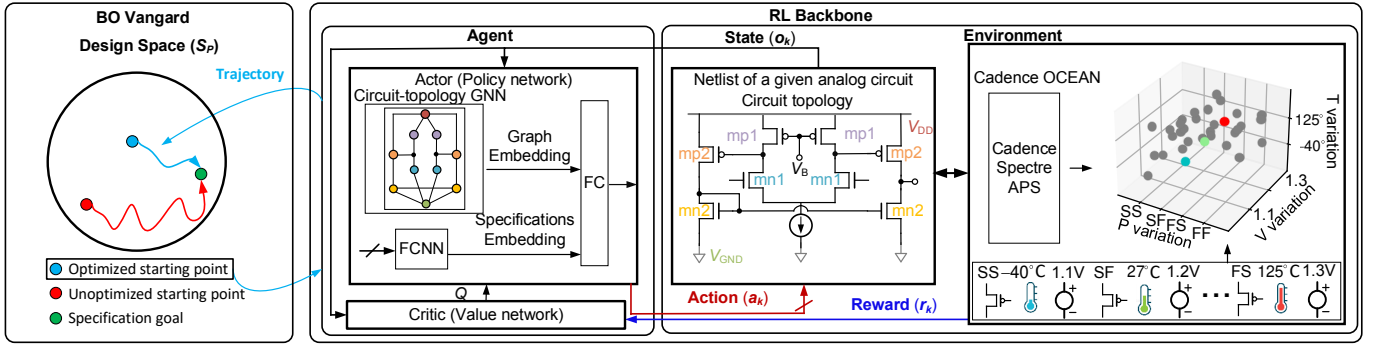


Fig. 1: Overview of the proposed RoSE method for efficient and robust analog sizing by incorporating BO Vanguard and RL Backbone.

used by the RL agent during both the training and deployment stage. Nonetheless, under the context of robust device sizing, both designs of these two parts are non-trivial and elaborated below.

1) *BO Vanguard*: We rely on BO to find an optimized initial searching point for our RL agent to improve its sampling efficiency during training. A key question that should be answered is what an optimized starting point is for our RL agent such that it does not particularly expedite the design for one specifically given group of design goals but generally facilitates the design of an arbitrarily given group of design goals from the whole sampling space S_G . Our idea is that from this starting point, the RL agent should averagely take the least deployment steps to achieve a general design. We thus let the device parameters found by BO that achieve as closely as possible the arithmetic mean of the maximum/minimum of each design goal in the whole sampling space S_G , i.e., $(PM_{\max} + PM_{\min})/2$, $(G_{\max} + G_{\min})/2$, $(BW_{\max} + BW_{\min})/2$, and $(P_{\max} + P_{\min})/2$, be the starting point for our RL agent.

We use a typical set-up of BO to search for the optimized starting point. BO is one of the surrogate-model-based black-box optimization methods including two essential parts: the surrogate model and the acquisition function [4]. The whole optimization depends on how accurately the surrogate model estimates the black-box function. Particularly, we adopt the widely used Gaussian process model as our surrogate model to predict the underlying black function with uncertainty. We use a Monte-Carlo-based Expected Improvement (EI) acquisition function to balance exploration and exploitation during the optimization by offering the next sampling point as below:

$$EI(X) \approx \frac{1}{Z} \sum_{i=1}^Z \max_{j=1, \dots, n} \{ \max(\xi_{ij} - f(s, g)_{best}, 0) \}, \quad (2)$$

$$\xi_i \sim \mathbb{P}(f(X) | \mathcal{D}).$$

Here, the expectation, $EI(X)$, is computed by approximating the integrals over the posterior distribution over Z points using Monte-Carlo sampling. $\mathbb{P}(f(X) | \mathcal{D})$ is the posterior distribution of the our function $f(s, g)$ at X where $X = (x_1, \dots, x_n)$ from the sampling in our design space S_P . $\mathcal{D}$ is our data set. Parameter ξ determines the amount of exploration during optimization. Note that we run our BO Vanguard multiple times with limited (50) simulations total to minimize the effect from the initial random sampling such that we can get a reliable optimized starting point.

2) *RL Backbone*: We rely on RL to tackle the complex device parameter optimization against PVT variations. Prior RL methods [7]–[10] for nominal case design are not readily

applicable in our framework. We thus non-trivially tailor an RL backbone for robust sizing by incorporating PVT variations into the design of its essential components, i.e., reward, state, action, agent, and environment. Here, we propose several techniques to design some critical parts.

Variation-aware reward: we connect the objective in Eq. (1) to our reward function so that our RL agent can directly optimize it by considering PVT variations. Particularly, the reward r_k at each time step k is designed by taking in PVT variations, i.e.,

$$r_k = \text{Mean} \left(\sum_{j=0}^{M-1} r_j \right); \text{ if } \exists j \in [0, M-1], r_j < 0; \quad (3)$$

$$\text{or } r_k = R, \text{ if } \forall j \in [0, M-1], r_j = 0.$$

Here, $r_j = \sum_{i=0}^{N-1} \min \{ \{s_i^j - g_i\} / \{s_i^j + g_i\}, 0 \}$ is the sub-reward of the j th corner, calculated by a weighted sum of the normalized difference between the intermediate circuit specification s_i^j and the given design goal g_i . All circuit specification types are equally important. M represents the number of PVT corners and N indicates the number of circuit specifications. In order not to overly optimize the specification, we set the upper bound of r_j to be 0. Only when circuit specifications under all PVT corners meet the design goal, a large stimulate reward of $R = 10$ is then given to encourage the agent for the successful design. Otherwise, the reward in each time step is the average of sub-rewards of all PVT corners. Finally, the accumulative reward for a training episode is $r_{episode} = \sum_{t=1}^{k=T} r_k$, where T is a pre-defined maximum step for an episode.

Domain-knowledge-based state: RL belongs to representation learning. To train a robust RL agent, capturing sufficient state (observation) from the environment with PVT variations is very important. We leverage the domain knowledge of analog circuits, i.e., intermediate circuit specifications and device parameters as our state, which covers most essential observations from a circuit design environment. Particularly, we take care of intermediate circuit specifications under all PVT corners in contrast to the prior work [11] which only considers partial PVT corners. The intermediate device parameters reflect the circuit's physical information. To better use them as observations, we distinguish different device types such as transistors, capacitors, and resistors with binary encoding. For example, if there are 4 different devices (e.g., N-type transistor, P-type transistor, capacitor, and resistor) in a given circuit, the N-type transistor can be identified as $[0, 0]$.

Circuit-aware policy: we adopt a critic (value network)-actor (policy network) architecture [9] to design our RL agent. The

TABLE I: Design space, sampling space, and PVT corners for three benchmark circuits.

Circuit types	Two-stage Op-Amp	Folded-cascode Op-Amp	Nested Miller compensation Op-Amp
Technology	GlobalFoundries 130/65/28 nm		
16 PVT conditions	Process: {SS, SF, FS, FF}	Voltage: {1.1V, 1.3V}	Temperature: {-40°C, 125°C}
Design space	10 ¹⁰ values	10 ¹⁷ values	10 ³⁹ values
Width (nm)	mp1: [1250, 2250, 10] mp2: [450, 2450, 20] mn1&mn2: [160, 260, 1]	mp1: [1000, 10000, 200] mp2: [1000, 10000, 200] mn1&mn2&mn3: [160, 1000, 20] c: [0.1, 10.0, 0.2]	mp1&mp2&mp3&mn4: [10000, 50000, 1000] mp4: [50000, 250000, 10000] mn1&mn2&mn3: [2000, 20000, 1000] c1: [25.0, 50.0, 0.5] c2: [1.0, 25.0, 0.5]
Capacitance (pF)	C_L : 1	C_L : 1	C_L : 100
Sampling space: Gain (dB) / I (A)	[10, 15] / [10 ⁻³ , 10 ⁻²]	[20, 30] / [10 ⁻⁴ , 10 ⁻³]	[40, 45] / [10 ⁻² , 10 ⁻¹]
PM (°) / BW (Hz)	[55] / [10 ⁵ , 4 × 10 ⁵]	[85] / [4 × 10 ⁶ , 6 × 10 ⁶]	[55] / [1 × 10 ⁶ , 2 × 10 ⁶]

value network has the same structure as the policy network except that it only outputs a scalar value to judge the action quality from the policy network. We thus emphasize the design of the policy network here. To better leverage the observations introduced before for the policy training, we propose a circuit-aware multi-modal policy network for our RL agent. The rationale is attributed to the heterogeneity between the different parts of the observations. Particularly, a fully-connected neural network (FCNN) takes intermediate circuit specifications under different PVT corners as input which are scalar values and can be flattened into a vector. And a circuit-topology-based graph neural network (GNN) is used to include the intermediate device parameters which have practical significance only when a detailed circuit topology is associated with them. The design details of this tailored GNN follow the same procedures as the original GNN work [13], which is omitted here. Combining the embeddings from these two parts using a fully connected (FC) layer forms the proposed circuit-aware multi-model policy network (Agent subset in Fig. 1), which can sufficiently leverage the domain-knowledge-based observations from PVT variation-aware environment for policy learning, thereby achieving robust sizing.

Similar to prior works [7], [9], we use discrete action space (i.e., increasing, decreasing, or remaining) to update device parameters with a predefined fine-grained step size. The design environment is in Cadence Spectre. Prior works [11] assume that the circuit simulation time scales linearly with the number of simulations, i.e., the simulation time for 16 PVT corners is 16× of the one of a single PVT corner. We use Cadence Spectre Accelerated Parallel Simulator (APS) to accelerate our simulation. At each time step, we can obtain circuit specifications for all PVT corners. Compared to a single PVT corner time, the batch simulation manner for 16 PVT corners only brings 0.17× time overhead as compared to one single PVT simulation. In other words, our circuit environment can extra achieve at most 14× sampling efficiency when collecting data points during training compared to prior RL methods. With this co-design loop, we are able to simultaneously achieve both high sampling efficiency and robust design by taking the advantage of BO, RL, and the simulation environment.

IV. EXPERIMENTS AND EVALUATIONS

A. Experiment Methodology

1) **Circuit Benchmarks:** Operational amplifiers (Op-Amps) are used as circuit benchmarks for evaluation because they are commonly used by prior arts [5]–[11]. Particularly, we adopt a two-stage Op-Amp, a Folded-cascode Op-Amp [14] and a three-stage nested Miller compensation Op-Amp with feedforward G_m stage [15] (NMCF). The design space for the device parameters, the sampling space for circuit specifications, and the PVT corners are listed in Table I. There

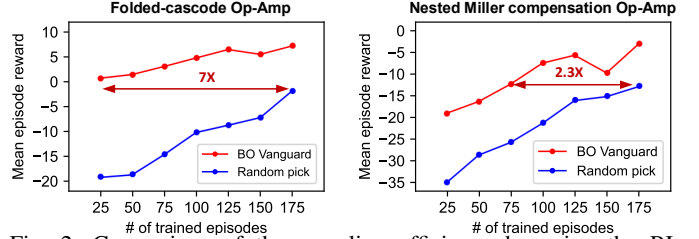


Fig. 2: Comparison of the sampling efficiency by using the RL Backbone with or without an optimized starting point as an example.

are $4 \times 2 \times 2 = 16$ extreme PVT conditions, including 4 process variations, 2 voltage variations, and 2 temperature variations.

2) **Metrics for Performance Evaluation:** We first show the sampling efficiency of RoSE by using a control experiment with/without the BO vanguard. The sampling efficiency is defined as the number of data points saved to achieve the same training quality (i.e., reward) as compared to another method. Following this evaluation, we then show the robustness of our method against PVT variations. Thirdly, we do an ablation study to distinguish the performance of RoSE with BO or RL. Finally, we compare our work to RobustAnalog–the prior RL work [11] targeting the same problem as us. Although there are several major differences, we care most about its efficacy in robust design as it uses task pruning with reduced PVT corners for RL training while our RL Backbone considers all PVT corners. Therefore, we follow this strategy to implement RobustAnalog by modifying our RL backbone, which in fact is an improved version of the original method as it also leverages several techniques proposed by us. RobustAnalog is only RL-based, we thus train our RL backbone without using BO vanguard. Additionally, to enable a reasonable comparison among different methods, we also propose a figure-of-merit (FoM) defined by using design accuracy and design efficiency: $\text{FoM} = \frac{\text{DA}}{N_D \cdot T_{\text{sim}}}$. Here, DA is the design accuracy (e.g., the deployment success rate to achieve 200 groups of design goals). N_D is the average number of required deployment steps (i.e., the number of circuit-level simulations) to reach a given group of design goals. T_{sim} is the simulation time for each circuit-level simulation run. $N_D \cdot T_{\text{sim}}$ indicates the design efficiency. Note that the training time for learning-based methods is not included here as it can be amortized during the inference stage once models are well trained.

We build the circuit-aware graph policy network using Deep Graph Library [16] and implement all methods with PyTorch and BoTorch [17]. We then use Proximal Policy Optimization (PPO) [18] to train RL agents. We use an equal amount of network parameters and the same set-ups for each baseline. We use Cadence Spectre as our simulator. All our experiments are performed on a 16-core Intel CPU. We use 65nm as a main example to show our results.

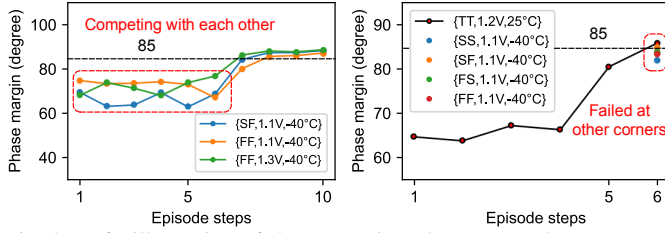


Fig. 3: Left: illustration of the competing phenomenon between PVT corners. Right: failed design by using the deployment with a single PVT corner at the nominal case. The dashed horizontal line is the design goal.

B. Evaluation Demonstration

1) **Efficient sampling with BO Vanguard:** we first show that the BO vanguard can indeed improve the sampling efficiency of our RL backbone. Fig. 2 illustrates the training curves of our RL backbone to design two types of Op-Amps with two different starting points, one is from BO searching (labeled as “BO Vanguard”) and the other one is a randomly picked value from the device parameter design space S_P , e.g., median value (labeled as “Random pick”). It shows that the RL agent without an optimized starting point often needs several more times training episodes to achieve the same reward as the one with an optimized starting point from BO (e.g., in this case, $7\times$ for Folded-cascode Op-Amp and $2.3\times$ for three-stage nested Miller compensation Op-Amp). Therefore, by optimizing the starting point, the RL agent converges faster with less sampling data (i.e., the circuit-level simulation runs).

2) **Robust design with the RL Backbone:** we then show the robust design enabled by our RL backbone against PVT variations by deploying our RL agent in an environment with full consideration of PVT variations. Policy deployment applies a trained policy to automatically find the device parameters for given design goals. The left column of Fig. 3 shows the deployment trajectories under several representative PVT corners by taking the phase margin of the Folded-cascode Op-Amp as an example, where each color represents a PVT corner. It can be seen that although each trajectory under a specific PVT corner is smooth, the worst corner can be quickly replaced by another corner due to the competition between different corners. Here, the worst case indicates the corner where the circuit specification deviates the most from the design goal as compared to all other corners. This phenomenon shows that device sizing with PVT variations is much more complex compared to the nominal case. Fortunately, by incorporating PVT variations into our method, our RL agent is able to achieve a robust design by finding optimal device parameters that can satisfy the design goal under all PVT corners. For further verification, we did a control experiment. We first find a group of optimal device parameters by deploying our trained RL agent at the nominal corner environment. We then verify the circuit specification under all other PVT corners by using the found device parameters. The right column of Fig. 3 illustrates such an example. It shows that the optimal device parameters obtained from a single PVT corner often do not satisfy all corners.

3) **Ablation study:** we then do an ablation study to investigate the performance of each part (i.e., BO Vanguard and RL Backbone) of our RoSE framework in device sizing with the consideration of PVT variations. Fig. 4 illustrates the mean deployment episode length (or design step) and mean deployment accuracy (or design accuracy) curves (or points) for RoSE, RL Backbone (or BO Vanguard). Here, for each sub-figure in the

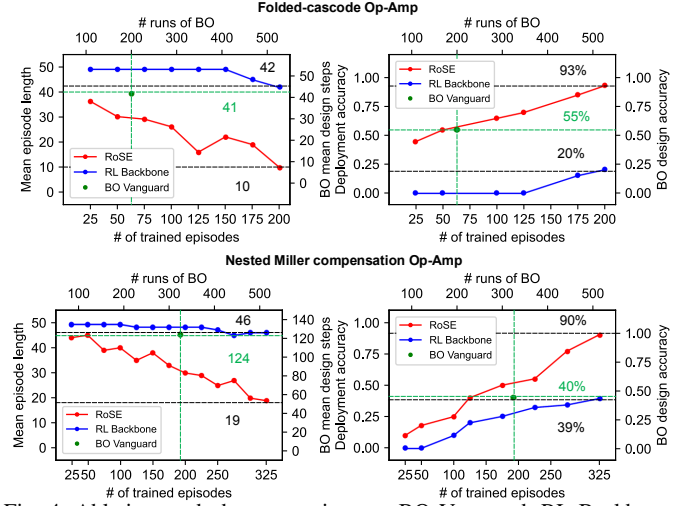


Fig. 4: Ablation study by comparing our BO Vanguard, RL Backbone, and their combination, RoSE, in terms of training sampling efficiency, deployment efficiency, and accuracy. Two rows correspond to the Folded-cascode Op-Amp [14] and the nested Miller compensation Op-Amp with feedforward G_m stage [15] (NMCF).

first column, the x -axis on the bottom is the number of trained episodes corresponding to RoSE and the RL Backbone, both of which requires training while the x -axis on the top is the number of runs of BO Vanguard starting from scratch to achieve given design goals. The y -axis on the left is the mean deployment episode length for RoSE and the RL Backbone while the y -axis on the right is the average number of design steps of each run used by BO Vanguard to search for optimal device parameters for a given group of design goals. Similarly, the axis of each sub-figure in the second column can be interpreted in the same way. Since BO Vanguard does not need training, its design step is marked by the solid green dot on the cross point of the vertical and the horizontal dashed line. The result shows that for the design of both Op-Amps, RoSE achieves the highest design efficiency (i.e, the least deployment steps, e.g., 10 vs. 42/41, to reach a group of given design goals) and the highest design accuracy (the highest success rate of deployment for given design goals, e.g., 93% vs. 20%/55% with the same number of episodes/runs). The comparison suggests that by combining BO and RL, RoSE can realize better performance than each of them.

4) **Full vs. partial PVT variation consideration:** we then further compare our method with RobustAnalog to show the advantage of full PVT variation consideration over partial PVT variation consideration. In this comparison, both methods use the same randomly-sampled starting point without BO for pre-optimization. Fig. 5 shows the training curves (i.e., mean episode length, and deployment accuracy) of two methods by using the design of the two-stage Op-Amp as an example. The maximum episode length for the RL agent is set to be 50. The total episodes used to train the two RL agents are chosen to be 600. Our RL Backbone realizes a higher design success rate (98% vs. 94%, or achieves the same design accuracy with $2\times$ less training epochs) and a shorter mean episode length (12.2) compared to the baseline (16.8). A shorter mean episode length means fewer steps are used to achieve the design goal, thereby achieving higher design efficiency. Each point in the deployment trajectory comes from deploying each RL agent for 200 groups of randomly sampled specifications in Table I.

TABLE II: Summary of Comparison with Existing Optimization Methods.

Methods	PVT incorporation	Optimized starting point	Sampling efficiency for policy training		FoM for policy deployment	
			folded-cascade	NMCF	folded-cascade	NMCF
RobustAnalog ^a [11]	Partial	No	1×	1×	0.33	0.59
BO Vanguard ^a	Full	N/A ^c	N/A ^c	N/A ^c	1.15	0.28
RL Backbone ^a	Full	No	2×	1.25×	0.41	0.72
RoSE^a	Full	Yes	>16×	3.25×	7.95	4.04
Bayesian Optimization ^b [4]	No	N/A ^c	N/A ^c	N/A ^c	0	0
RL baseline ^b [9]	No	No	Fail	Fail	0	0

^a RobustAnalog, BO Vanguard, RL Backbone, and RoSE consider PVT variations corners during training/optimization.

^b Representatives of prior arts [2]–[10] that ignore PVT variations in training/optimization. They are implemented by our BO and RL without taking in PVT variations.

^c BO methods do not need optimized starting points and training. Therefore, some metrics are not applicable here.

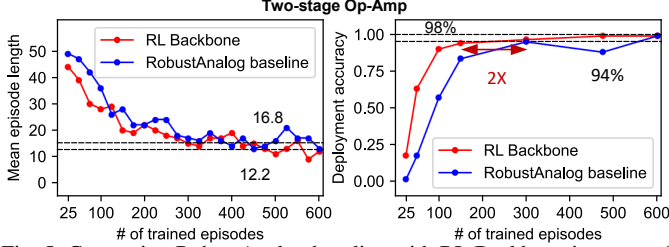


Fig. 5: Comparing RobustAnalog baseline with RL Backbone in terms of training sampling efficiency, deployment efficiency, and design accuracy.

Although here the RobustAnalog is an advanced implementation of the original one [11], the comparison shows that our method with full consideration of PVT variations is still superior to it.

5) **Comparison Summary:** finally, we summarize the features of RoSE together with the above comparisons in Table II. Here, we also add some other comparisons in this table, such as directly applying prior arts [2]–[10] which ignore PVT variations to do device sizing in an environment with PVT variations. Here, we use Bayesian Optimization [4] and RL [9] as the representatives of the prior arts, i.e., optimization-based [2]–[4] and learning-based methods [5]–[10]. Additionally, we use the FoM defined before to show the overall performance of a design automation method in robust device sizing. Without considering PVT variations indeed fail in the robust design with zero design accuracy. The comparisons show that with BO as our pre-optimization, our RoSE framework can significantly improve sampling efficiency, design efficiency, and design accuracy with regard to the PVT-aware design. Note that BO is only performed once at the very beginning of our framework, thereby incurring minimal overhead. In summary, our proposed RoSE method that leverages the complementary advantages of BO and RL can achieve the best FoM for the challenging robust device sizing problem.

V. CONCLUSION

We have presented an efficient and robust sizing method RoSE for automated analog circuit parameter optimization based on BO and RL. We take the advantage of BO’s fast convergence to find an optimized starting point for RL to achieve efficient training and deployment. We then leverage the RL method’s strong optimization ability by incorporating sufficient PVT variations into the framework through the tailored circuit-aware multi-modal policy network and customized variation-aware reward function. In this way, the RL can effectively learn a robust policy for device parameter optimization. Evaluations demonstrate our RoSE’s superior performance as compared to state-of-the-art works.

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